Manufacturer

ACME Metal Works GmbH

Industrial Park 123
52066Aachen
DE
quality@acme-metal.example.com

Customer

Engineering Solutions Ltd.

Tech Park Way 45
Cardiff
CF14 5DU
GB
procurement@engisolutions.example.com

Subcustomer

American Heavy Industries Inc.

5000 Industrial Blvd
Building C
Detroit, MI48201
US
materials@heavyind.example.com

Goods Receiver

Canadian Construction Corp

777 Construction Ave
Toronto, ONM5H 2N2
CA
logistics@canconstruct.example.ca

Digital Material Passport

ID	DMP-METAL-002	*Version*	1.0.0
Issue Date	2025-05-14	*Certificate Type*	EN 10204 3.1

Business Transaction

Order		**Delivery**	
Order ID	PO-78902	*Delivery ID*	DN-56790
Position	10	*Position*	1
Date	2025-04-21	*Date*	2025-05-13
Quantity	2000 kg	*Quantity*	2000 kg

Product Information

Product Name	Structural Steel S420N
Batch ID	H-10988-01
Surface Condition	Hot-rolled
Weight	2000 kg
Production Date	2025-05-10
Country of Origin	DE
Flatness Tolerance	Class N - Normal

Product Norms

Standard	EN 10025-3 (2019)

Material Designations

Name (EN)	1.8902

Product Shape

Form	Plate
Length	6000 mm
Width	2000 mm
Thickness	25 mm

Heat Treatment

Process	*Lot*	*Furnace*	*Date*
Normalizing	NORM-2024-0823-A12	NF-LINE-02	2024-08-23

Stages

Stage	*Temperature*	*Duration*	*Cooling*	*Atmosphere*
Austenitizing	920 C	60 min		Air
Cooling	20 C		Air	

Chemical Analysis

Heat Number	H-10988
Melting Process	EAF+LF+VD
Casting Date	2025-05-09
Casting Method	ContinuousCasting
Sample Location	Ladle

Elements

Symbol	**C**	**Mn**	**Si**	**P**	**S**	**CEV**
Unit	%	%	%	%	%	%
Min	-	-	-	-	-	-
Max	0.2	1.6	0.5	0.025	0.015	0.44
Actual	0.16	1.48	0.28	0.016	0.01	0.41

Formula Definitions

CEV = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15: 0.41%

Mechanical Properties

Property	*Symbol*	*Actual*	*Minimum*	*Maximum*	*Method*	*Status*
Tensile Strength Test temperature: 20°C Specimen: 1/3R, L					EN ISO 6892-1	✓

Individual Values	# 1	# 2	# 3
Value [MPa]	558	562	560

Statistics	**Mean**	**Min/Max**	**Std Dev**
EN ISO 6892-1 statistical analysis	560	558 / 562	2 (Sample)

Property	*Symbol*	*Actual*	*Minimum*	*Maximum*	*Method*	*Status*
Yield Strength Test temperature: 20°C, 3 specimens tested					EN ISO 6892-1	✓

Individual Values	# 1	# 2	# 3
Value [MPa]	442	445	448

Statistics	**Mean**	**Min/Max**	**Std Dev**
EN ISO 6892-1 statistical analysis	445	442 / 448	3 (Sample)

Property	*Symbol*	*Actual*	*Minimum*	*Maximum*	*Method*	*Status*
Elongation after fracture Gauge length 5.65#S_0, 3 specimens tested					EN ISO 6892-1	✓

Individual Values	# 1	# 2	# 3
Value [%]	23	24	25

Statistics	**Mean**	**Min/Max**	**Std Dev**
EN ISO 6892-1 statistical analysis	24	23 / 25	1 (Sample)

Reduction of Area EN ISO 6892-1 ✓
3 specimens tested

Individual Values	# 1	# 2	# 3
Value [%]	60	62	64

Statistics	Mean	Min/Max	Std Dev
EN ISO 6892-1 statistical analysis	62	60 / 64	2 (Sample)

Charpy V-notch Impact Energy EN ISO 148-1 ✓
Test temperature: -20°C
Specimen: Top, T-L

Individual Values	# 1	# 2	# 3
Value [J]	56	58	60

Statistics	Mean	Min/Max	Std Dev
EN ISO 148-1 statistical analysis	58	56 / 60	2 (Sample)

Charpy V-notch Impact Energy EN ISO 148-1 ✓
Test temperature: -20°C
Specimen: Bottom, T-L

Individual Values	# 1	# 2	# 3
Value [J]	54	57	59

Statistics	Mean	Min/Max	Std Dev
EN ISO 148-1 statistical analysis	56.7	54 / 59	2.5 (Sample)

Brinell Hardness EN ISO 6506-1 ✓
Ball: 10mm, Force: 3000kg, 5 indentations

Individual Values	# 1	# 2	# 3	# 4	# 5
Value [HBW]	183	185	187	184	186

Statistics	Mean	Min/Max	Std Dev
EN ISO 6506-1 statistical analysis	185	183 / 187	1.58 (Sample)

Vickers Hardness EN ISO 6507-1 ✓
10 kg load, 5 indentations

Individual Values	# 1	# 2	# 3	# 4	# 5
Value [HV10]	192	195	198	194	196

Statistics	Mean	Min/Max	Std Dev
EN ISO 6507-1 statistical analysis	195	192 / 198	2.35 (Sample)

Rockwell Hardness	HR	18HRC		22HRC	EN ISO 6508-1	✓
Elastic Modulus	E	210GPa			EN ISO 6892-1	✓
Strain Hardening - Exponent	n	0.18			ASTM E646	✓
Plastic Strain Ratio	r	1.2	1		EN ISO 10113	✓

0.2% Proof Strength EN ISO 6892-1 ✓
3 specimens tested

Individual Values	# 1	# 2	# 3
Value [MPa]	428	430	432

Statistics	Mean	Min/Max	Std Dev
EN ISO 6892-1 statistical analysis	430	428 / 432	2 (Sample)

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with the requirements of EN 10204:2004 type 3.1 and the specified standards. The results comply with the requirements.

Validated By

Name	*Title*	*Department*	*Date*
Johann Weber	Quality Inspector	Quality Assurance	2025-05-14

Data schema maintained by Material Identity. https://schemas.materialidentity.org/metals-schemas/v0.1.1/schema.json